

Lecture 3: Physical Layer - Guided Media and Digital Signaling

EC 441 - Introduction to Computer Networking

Boston University, Spring 2026

Learning Objectives

By the end of this lecture, you should be able to:

1. Describe the characteristics of guided transmission media (twisted pair, coaxial, fiber)
2. Calculate signal attenuation using dB/km models
3. Understand the relationship between bandwidth and data rate
4. Model a communication channel using the AWGN model
5. Analyze digital signaling schemes (unipolar, bipolar, multi-level)
6. Design and analyze matched filter receivers
7. Calculate probability of error using the Q-function
8. Understand the tradeoffs between power, bandwidth, error rate, and data rate

1 Physical Transmission Media

1.1 Why Study Physical Media?

Networks ultimately transmit **electromagnetic energy**:

- Electrical signals on copper wires
- Light pulses in optical fibers
- Radio waves through air

Understanding physical constraints is critical for:

- **Link budget analysis** (can signal reach destination?)
- **Data rate limits** (how fast can we transmit?)
- **Network design** (which medium for which application?)

Medium	Bandwidth	Distance	Cost	Use Case
Twisted Pair	1-1000 MHz	100 m	Low	Ethernet, phone
Coaxial Cable	1-1000 MHz	500 m	Medium	Cable TV, legacy
Optical Fiber	>10 THz	100 km+	High	Long-haul, backbone

Table 1: Comparison of guided transmission media

1.2 Guided Media Overview

1.3 Twisted Pair Cable

Structure:

- Two insulated copper wires twisted together
- Twisting reduces electromagnetic interference (EMI)
- Multiple pairs bundled in a cable

Categories (Ethernet):

- **Cat 5e:** 100 MHz, up to 1 Gbps
- **Cat 6:** 250 MHz, up to 10 Gbps (short distance)
- **Cat 6a:** 500 MHz, 10 Gbps up to 100 m
- **Cat 8:** 2000 MHz, 25-40 Gbps (data centers)

Key Parameters:

Characteristic Impedance: $Z_0 = 100 \Omega$ (for Ethernet)

- Must match transmitter and receiver impedance to avoid reflections

Attenuation: $\sim 2\text{-}3 \text{ dB per } 100\text{m at } 100 \text{ MHz}$

- Increases with frequency: $\alpha \propto \sqrt{f}$

Propagation speed: $v \approx 0.64c$ (where c = speed of light)

- Dielectric slows signal compared to free space

Twisted Pair Attenuation

For Cat 6 cable at 100 MHz over 100m:

```

Attenuation: ~2.5 dB/100m
If transmit power: P_tx = 1 W = 30 dBm
Received power: P_rx = 30 dBm - 2.5 dB = 27.5 dBm ~ 0.56 W
Power loss: 44%
```

1.4 Coaxial Cable

Structure:

- Central conductor (copper wire)
- Dielectric insulator
- Outer conductor (shield/braid)
- Protective jacket

Advantages:

- Better shielding than twisted pair
- Higher bandwidth capability
- More consistent impedance

Types:

- **RG-6** (75 Ω): Cable TV, cable modem
- **RG-58** (50 Ω): Legacy 10Base2 Ethernet
- **Hardline**: High-power applications

Key Parameters:

Characteristic Impedance:

$$Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln \left(\frac{D}{d} \right) \quad (1)$$

where D = outer conductor diameter, d = inner conductor diameter, ϵ_r = relative permittivity

Attenuation: 0.5-2 dB per 100m (lower than twisted pair)

Bandwidth: DC to 1+ GHz (depending on type)

Modern Use:

- Cable TV distribution (DOCSIS)
- Cable Internet: up to 10 Gbps (DOCSIS 3.1)
- Less common for new LANs (replaced by twisted pair and fiber)

1.5 Optical Fiber

Basic Principle:

- Light propagates through glass core via **total internal reflection**
- Core has higher refractive index than cladding
- Critical angle: $\theta_c = \arcsin(n_2/n_1)$

Types:

Single-Mode Fiber (SMF):

- Core diameter: $\sim 9 \mu\text{m}$
- One light path (mode)
- Very low dispersion
- Distance: 100+ km
- Use: Long-haul, backbone

Multi-Mode Fiber (MMF):

- Core diameter: 50-62.5 μm
- Multiple light paths
- Higher dispersion (limits distance)
- Distance: 2 km typical
- Use: Data centers, campus networks

Key Parameters:

Attenuation:

- 1550 nm wavelength: $\sim 0.2 \text{ dB/km}$ (ultra-low!)
- 1310 nm wavelength: $\sim 0.4 \text{ dB/km}$
- 850 nm wavelength (MMF): $\sim 2.5 \text{ dB/km}$

Compare to copper at 100 MHz: $\sim 25 \text{ dB/km} \rightarrow$ fiber has **100 $\times$ less attenuation**

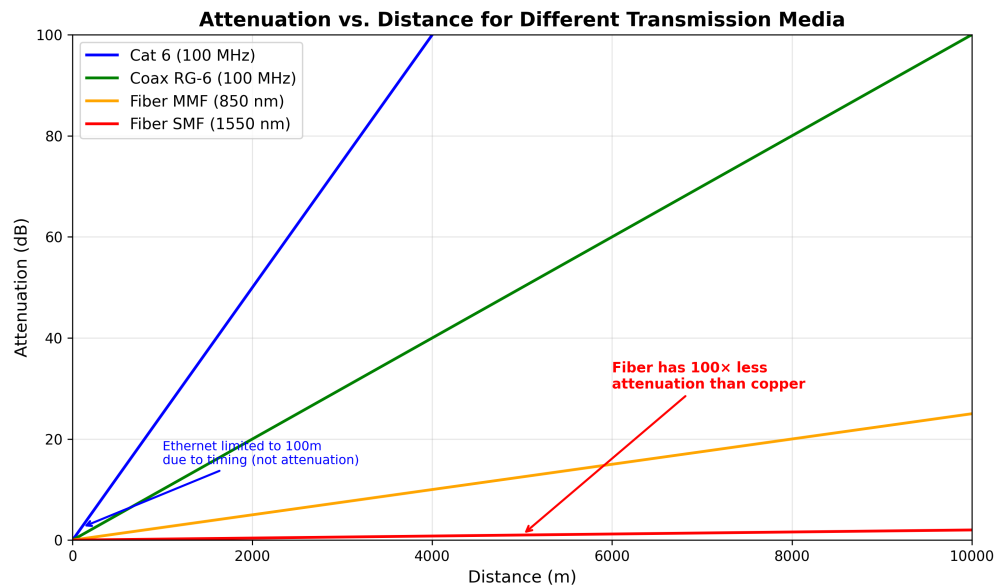


Figure 1: Attenuation comparison across different cable types and frequencies. Optical fiber shows dramatically lower attenuation than copper-based media.

Bandwidth: Enormous!

- Single fiber: >10 THz of spectrum
- WDM (wavelength division multiplexing): multiple channels
- Modern systems: 100+ wavelengths $\times$ 100 Gbps = 10+ Tbps per fiber

Propagation speed: $v \approx 0.67c$ (in glass)

- Refractive index $n \approx 1.5$
- Speed: $v = c/n$

Dispersion: Limits data rate $\times$ distance product

- **Chromatic dispersion:** Different wavelengths travel at different speeds
- **Modal dispersion:** Different modes travel different paths (MMF only)
- Typical: ~ 17 ps/(nm·km) at 1550 nm

Long-Haul Fiber Link

100 km fiber link at 1550 nm:

```
Attenuation: 0.2 dB/km x 100 km = 20 dB
If P_tx = 1 mW = 0 dBm
Then P_rx = 0 dBm - 20 dB = -20 dBm = 10 uW
Power ratio: 100:1 loss (still plenty of signal!)
```

1.6 Attenuation in Decibels (dB)

Why use dB? Multiplication becomes addition!

Power ratio in dB:

$$\text{dB} = 10 \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right) \quad (2)$$

Absolute power levels:

dBm (relative to 1 mW):

$$\text{dBm} = 10 \log_{10} \left(\frac{P}{1 \text{ mW}} \right) \quad (3)$$

dBW (relative to 1 W):

$$\text{dBW} = 10 \log_{10} \left(\frac{P}{1 \text{ W}} \right) \quad (4)$$

Common Values to Memorize:

Link Budget Calculation:

$$P_{\text{rx}}(\text{dBm}) = P_{\text{tx}}(\text{dBm}) - L_{\text{cable}}(\text{dB}) + G_{\text{tx}}(\text{dBi}) + G_{\text{rx}}(\text{dBi}) \quad (5)$$

Power Ratio	dB	Description
2	3 dB	double
10	10 dB	-
100	20 dB	-
1/2	-3 dB	half
1/10	-10 dB	-

1.7 Bandwidth and Data Rate Relationship

Shannon-Hartley Theorem:[1]

$$C = B \log_2 \left(1 + \frac{S}{N} \right) \text{ bits/s} \quad (6)$$

where:

- C = channel capacity (maximum achievable data rate)
- B = bandwidth (Hz)
- S/N = signal-to-noise ratio (linear, not dB)

Key Insights:

1. Capacity increases logarithmically with SNR

- Doubling SNR doesn't double capacity
- High SNR $\rightarrow$ diminishing returns

2. Capacity increases linearly with bandwidth

- Doubling bandwidth doubles capacity
- **Bandwidth is precious!**

3. Can trade SNR for bandwidth

- Low bandwidth + high power
- High bandwidth + low power

Shannon Capacity Calculations

Telephone channel: $B = 3 \text{ kHz}$, $\text{SNR} = 30 \text{ dB}$ (1000:1)

$$C = 3000 \times \log_2(1 + 1000) \approx 3000 \times 10 = 30 \text{ kbps}$$

Ethernet link: $B = 100 \text{ MHz}$, $\text{SNR} = 20 \text{ dB}$ (100:1)

$$C = 10^8 \times \log_2(101) \approx 10^8 \times 6.66 = 666 \text{ Mbps}$$

Note: Shannon capacity is a theoretical limit. Practical systems achieve 50-90% of capacity.

1.8 Propagation Speed

Speed of electromagnetic waves in medium:

$$v = \frac{c}{\sqrt{\epsilon_r}} \quad (7)$$

where:

- $c = 3 \times 10^8$ m/s (speed of light in vacuum)
- ϵ_r = relative permittivity (dielectric constant)

Typical Values:

- **Vacuum:** $v = c$ (baseline)
- **Air:** $v \approx 0.997c$ (essentially speed of light)
- **Twisted pair:** $v \approx 0.64c$ (polyethylene insulation, $\epsilon_r \approx 2.3$)
- **Coax:** $v \approx 0.66c$ (similar dielectric)
- **Optical fiber:** $v \approx 0.67c$ (glass, $n \approx 1.5$)

Latency Calculation

New York to Los Angeles: ~4000 km

```
Fiber propagation time: 4000 km / (0.67 x 3x108 km/s) = 20 ms
Round-trip time: ~40 ms (propagation only)
Measured ping: ~60-80 ms (includes router processing, queuing)
```

Fundamental Limit:

- **Nothing** travels faster than light in vacuum
- Physical distance sets a hard lower bound on latency
- Why CDNs place servers geographically close to users!

2 Hands-On Activity 1: Link Budget Calculations

2.1 Problem 1: Ethernet Link

A 1000BASE-T Ethernet link transmits at 30 dBm over 100m of Cat 6 cable.

- Attenuation: 2.5 dB per 100m
- Receiver sensitivity: -30 dBm (minimum to decode signal)

Calculate:

1. Received power level
2. Link margin (excess signal above minimum)

3. Maximum cable length

Solution

1. $P_{\text{rx}} = 30 \text{ dBm} - 2.5 \text{ dB} = 27.5 \text{ dBm}$
2. $\text{Margin} = 27.5 - (-30) = 57.5 \text{ dB}$ (huge margin!)
3. Max length: $57.5 \text{ dB} / 2.5 \text{ dB per } 100\text{m} = 23 \times 100\text{m} = 2.3 \text{ km}$
(But Ethernet spec limits Cat 6 to 100m for timing reasons)

2.2 Problem 2: Fiber Optic Link

A 100 km fiber link at 1550 nm:

- Transmit power: 10 dBm
- Fiber loss: 0.2 dB/km
- Splice loss: 0.1 dB per splice (10 splices)
- Connector loss: 0.5 dB per connector (2 connectors)
- Receiver sensitivity: -25 dBm

Calculate:

1. Total fiber loss
2. Total loss (including splices and connectors)
3. Received power
4. Link margin

Solution

1. Fiber loss: $0.2 \times 100 = 20 \text{ dB}$
2. Splice loss: $0.1 \times 10 = 1 \text{ dB}$; Connector: $0.5 \times 2 = 1 \text{ dB}$
Total loss: $20 + 1 + 1 = 22 \text{ dB}$
3. $P_{\text{rx}} = 10 - 22 = -12 \text{ dBm}$
4. Margin: $-12 - (-25) = 13 \text{ dB}$ (good margin for aging)

3 AWGN Channel Model**3.1 Communication System Block Diagram**

A simplified model for the physical layer consists of:

- **Transmitter:** Generates signal $x(t)$

- **Channel:** Modifies signal (attenuation, delay)
- **Noise:** Adds random disturbance $n(t)$
- **Receiver:** Observes $y(t) = h(t) * x(t) + n(t)$

3.2 Additive White Gaussian Noise (AWGN) Channel

Mathematical Model:

$$y(t) = h(t) * x(t) + n(t) \quad (8)$$

where:

- $x(t)$ = transmitted signal
- $h(t)$ = channel impulse response
- $*$ = convolution operator
- $n(t)$ = noise process
- $y(t)$ = received signal

Noise Characteristics:

1. **Additive:** Noise adds to signal (doesn't multiply or distort)
2. **White:** Noise power is equal at all frequencies
 - Power spectral density: $N_0/2$ W/Hz (constant)
3. **Gaussian:** Noise amplitude follows Gaussian distribution
 - $n(t) \sim \mathcal{N}(0, \sigma^2)$
 - Justified by Central Limit Theorem (sum of many small noise sources)

Why AWGN?

- **Mathematically tractable** (we can analyze it!)
- **Good model for thermal noise** in receivers
- **Pessimistic** (real channels often better)
- **Fundamental limit** on performance

3.3 Sources of Noise

3.3.1 Thermal Noise (Johnson-Nyquist Noise)

Generated by random motion of electrons in conductors.[2]

Available noise power:

$$N = k_B T B \quad (9)$$

where:

- $k_B = 1.38 \times 10^{-23}$ J/K (Boltzmann constant)
- T = temperature (Kelvin)
- B = bandwidth (Hz)

Thermal Noise Power

Room temperature, 1 MHz bandwidth:

$$N = (1.38 \times 10^{-23}) \times 300 \times 10^6 = 4.14 \times 10^{-15} \text{ W} = -144 \text{ dBm}$$

Key insight: Thermal noise power is proportional to bandwidth!

- Wider bandwidth $\rightarrow$ more noise power
- Can't escape thermal noise (unless you cool to 0 K!)

3.3.2 Shot Noise

Due to discrete nature of electric charge.

- Important in photodetectors (optical receivers)
- Less significant in radio systems

3.3.3 Interference

Not truly “noise” but often modeled as such:

- Adjacent channel interference
- Co-channel interference
- Impulse noise (switching, lightning)

3.4 Noise Figure and Noise Temperature

Noise Figure (NF):[3]

Measures how much a component (amplifier, mixer) degrades SNR.

$$\text{NF} = 10 \log_{10} \left(\frac{\text{SNR}_{\text{in}}}{\text{SNR}_{\text{out}}} \right) \text{ dB} \quad (10)$$

Friis Formula (cascaded components):

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots \quad (11)$$

where F = noise factor (linear), G = gain (linear)

Key insight: First stage (LNA = low-noise amplifier) dominates!

- Keep first-stage NF as low as possible
- Subsequent stages matter less if first stage has high gain

Noise Temperature (T_e):

Alternative way to express noise figure:

$$T_e = T_0(F - 1) \quad (12)$$

where $T_0 = 290$ K (reference temperature)**Noise Temperature Calculation**NF = 3 dB $\rightarrow F = 2$ (linear) $\rightarrow T_e = 290$ KTotal noise temperature: $T_{\text{sys}} = T_{\text{antenna}} + T_e$ **3.5 Distortionless Channel Model****Ideal transmission:** Signal shape preserved, only scaled and delayed.

$$y(t) = K \cdot x(t - t_d) + n(t) \quad (13)$$

where:

- K = attenuation factor ($K < 1$)
- t_d = propagation delay

Frequency domain:

$$H(f) = K e^{-j2\pi f t_d} \quad (14)$$

- Constant magnitude: $|H(f)| = K$
- Linear phase: $\angle H(f) = -2\pi f t_d$

Real channels deviate from ideal:

- **Frequency-dependent attenuation:** High frequencies attenuate more
- **Dispersion:** Different frequencies travel at different speeds
- **Multipath:** Signal takes multiple paths (wireless, indoor)

For now, we focus on the idealized model with distance-based attenuation:

$$K = 10^{-\alpha d/20} \quad (15)$$

where α = attenuation in dB/km, d = distance in km**3.6 Signal-to-Noise Ratio (SNR)****Definition:**

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{S}{N} \quad (16)$$

In dB:

$$\text{SNR (dB)} = 10 \log_{10} \left(\frac{S}{N} \right) = P_s(\text{dBm}) - P_n(\text{dBm}) \quad (17)$$

Typical Values:

- Voice (acceptable): 10-15 dB
- Music (good quality): 40-60 dB
- Digital communication: 10-30 dB (depends on modulation)

SNR Calculation

Received signal: -70 dBm, Noise floor: -100 dBm

$$\text{SNR} = -70 - (-100) = 30 \text{ dB}$$

Convert to linear: $\text{SNR} = 10^{30/10} = 1000$

This is quite good for digital communication!

4 Digital Signaling Fundamentals

4.1 Representing Bits as Signals

Goal: Transmit digital data (bits) over physical channel (analog waveforms)

General form of transmitted signal:[4]

$$x(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT) \quad (18)$$

where:

- a_k = symbol values (determined by data bits)
- $g(t)$ = pulse shape (determines bandwidth)
- T = symbol period (symbol rate = $1/T$ symbols/sec)

Key Design Choices:

1. **Pulse shape** $g(t)$: rectangular, raised cosine, Gaussian, etc.
2. **Symbol mapping** a_k : how bits map to symbol values
3. **Symbol rate** $1/T$: related to bandwidth

4.2 Bit-to-Symbol Mapping

Binary Signaling (1 bit/symbol):

Unipolar (On-Off Keying, OOK):

- Bit 0: $a_k = 0$
- Bit 1: $a_k = A$
- Simple, but wastes energy (average power = $A^2/2$)
- Baseline wander in long strings of 0s

Polar (Bipolar) NRZ (Non-Return-to-Zero):

- Bit 0: $a_k = -A$
- Bit 1: $a_k = +A$
- Balanced (average value = 0, no DC component)
- Better noise immunity
- **Most common for baseband signaling**

Return-to-Zero (RZ):

- Bit 0 or 1: pulse returns to zero during symbol period
- Uses more bandwidth
- Easier clock recovery

4.3 Pulse Shapes**4.3.1 Rectangular Pulse**

$$g(t) = \begin{cases} 1 & 0 \leq t < T \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

- **Bandwidth:** $B \approx 1/T$ (sinc spectrum, main lobe)
- **Pros:** Simple to generate
- **Cons:** Infinite bandwidth (in theory), intersymbol interference (ISI)

4.3.2 Raised Cosine Pulse

[5]

Frequency domain representation:

$$G(f) = \begin{cases} T & |f| \leq \frac{1-\alpha}{2T} \\ \frac{T}{2} \left[1 + \cos \left(\frac{\pi T}{\alpha} \left(|f| - \frac{1-\alpha}{2T} \right) \right) \right] & \frac{1-\alpha}{2T} < |f| \leq \frac{1+\alpha}{2T} \\ 0 & |f| > \frac{1+\alpha}{2T} \end{cases} \quad (20)$$

where α = roll-off factor ($0 \leq \alpha \leq 1$)

- **Bandwidth:** $B = \frac{1+\alpha}{2T}$
- **Pros:** Controllable bandwidth, zero ISI at sampling points
- **Cons:** More complex to generate
- **Used in:** WiFi, cellular, satellite

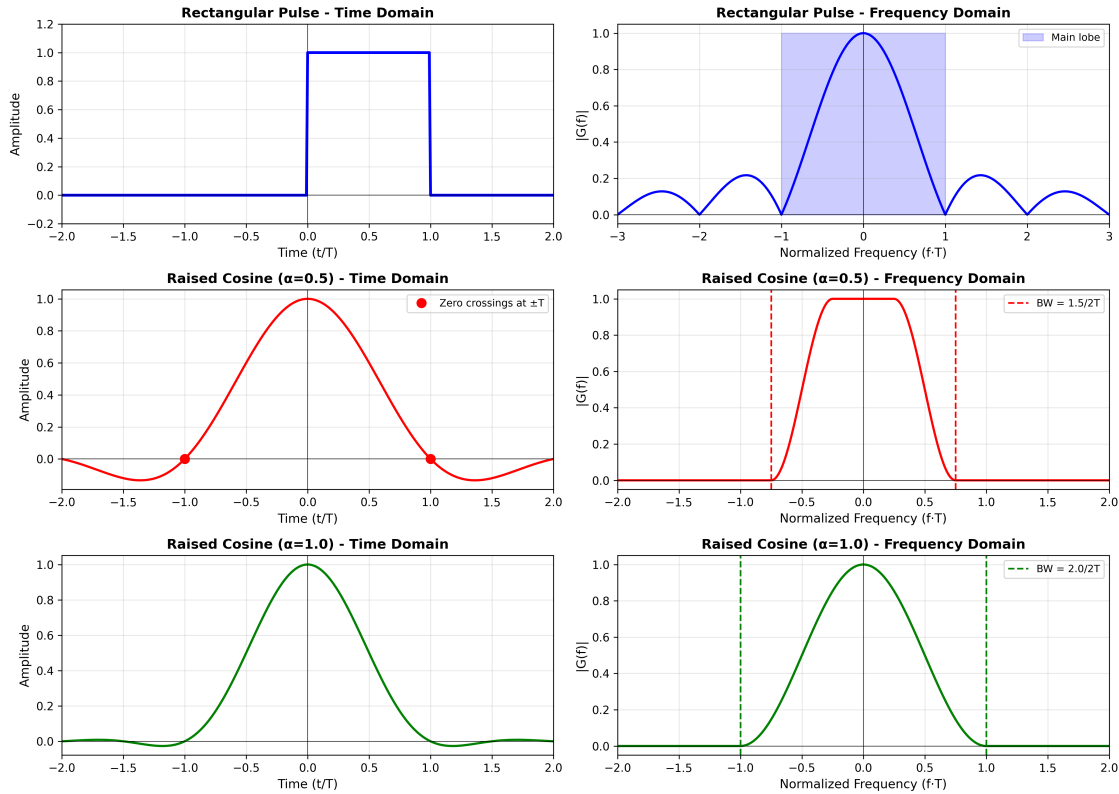


Figure 2: Pulse shapes in time and frequency domain. Top: rectangular, raised cosine with different roll-off factors. Bottom: corresponding frequency spectra showing bandwidth requirements.

4.4 Energy per Bit (E_b) and E_b/N_0

Energy per bit:

$$E_b = \int_0^T x^2(t) dt \quad (21)$$

For bipolar NRZ with amplitude A and rectangular pulse:

$$E_b = A^2 T \quad (22)$$

Noise power spectral density: $N_0/2$ (W/Hz, two-sided)

Key performance metric: E_b/N_0 (bit energy to noise density ratio)

$$\frac{E_b}{N_0} = \frac{\text{Energy per bit}}{\text{Noise spectral density}} \quad (23)$$

Why use E_b/N_0 instead of SNR?

- **Normalizes for data rate:** Different data rates can be compared
- **Independent of bandwidth:** Focuses on fundamental limits
- **Universal metric:** Used across all modulation schemes

Relationship to SNR:

$$\frac{E_b}{N_0} = \frac{S}{N} \cdot \frac{T}{1} = \frac{S}{N} \cdot \frac{B}{R_b} \quad (24)$$

where R_b = bit rate, B = bandwidth

For rectangular pulse, $B \approx R_b$, so:

$$\frac{E_b}{N_0} \approx \frac{S}{N} \quad (25)$$

5 Matched Filter Receiver

5.1 Optimal Receiver Design

Goal: Minimize probability of bit error in AWGN channel

Received signal:

$$y(t) = a_k g(t - kT) + n(t), \quad kT \leq t < (k+1)T \quad (26)$$

Matched Filter Theorem:[6]

To maximize SNR at the sampling instant, use a filter matched to the pulse shape:

$$h_{\text{MF}}(t) = g(T - t) \quad (27)$$

That is, $h_{\text{MF}}(t) = g(-t)$ time-reversed, sampled at $t = T$.

Receiver Structure:

The receiver consists of:

1. Matched filter $h(t) = g(T - t)$
2. Sample at $t = T$
3. Decision device

Output of matched filter at sampling time:

$$z(T) = \int_{-\infty}^{\infty} y(t)g(T - t) dt = a_k E_g + n_s \quad (28)$$

where:

- $E_g = \int_{-\infty}^{\infty} g^2(t) dt$ = pulse energy
- n_s = noise sample (Gaussian, variance $\sigma^2 = N_0 E_g / 2$)

Maximum SNR at sampler:

$$\text{SNR}_{\text{out}} = \frac{a_k^2 E_g}{N_0 / 2} \quad (29)$$

5.2 Decision Device (ML Detector)

For binary signaling ($a_k \in \{-A, +A\}$):

Decision rule: If $z > 0$, decide $\hat{a}_k = +A$ (bit 1); else $\hat{a}_k = -A$ (bit 0)

This is the **maximum likelihood (ML) decision** for equally likely bits.

Threshold:

- For symmetric signaling (bipolar): threshold = 0
- For asymmetric (e.g., OOK): threshold = $(A_0 + A_1)/2$

For M -ary signaling:

- Multiple decision regions
- Minimum distance criterion

5.3 Probability of Error

Received sample after matched filter:

$$z = a_k + n_s \quad (30)$$

where $n_s \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = N_0/2$

For bipolar signaling ($a_k = \pm A$):

Assume bit 1 sent ($a_k = +A$):

$$z = A + n_s \quad (31)$$

Error occurs if $z < 0$ (noise pushes sample below threshold):

$$P(\text{error}|a_k = +A) = P(n_s < -A) = Q\left(\frac{A}{\sigma}\right) \quad (32)$$

where $Q(x)$ is the **Q-function**: [7]

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du \quad (33)$$

Bit Error Rate (BER) for bipolar NRZ:

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \quad (34)$$

Key property: $Q(x)$ decreases **very rapidly** with x

- $Q(3) \approx 1.3 \times 10^{-3}$ (1 error per 1000 bits)
- $Q(4) \approx 3 \times 10^{-5}$ (1 error per 30,000 bits)
- $Q(5) \approx 3 \times 10^{-7}$ (1 error per 3 million bits)
- $Q(6) \approx 10^{-9}$ (1 error per billion bits)

This is a **waterfall curve** - small increase in SNR gives huge decrease in BER!

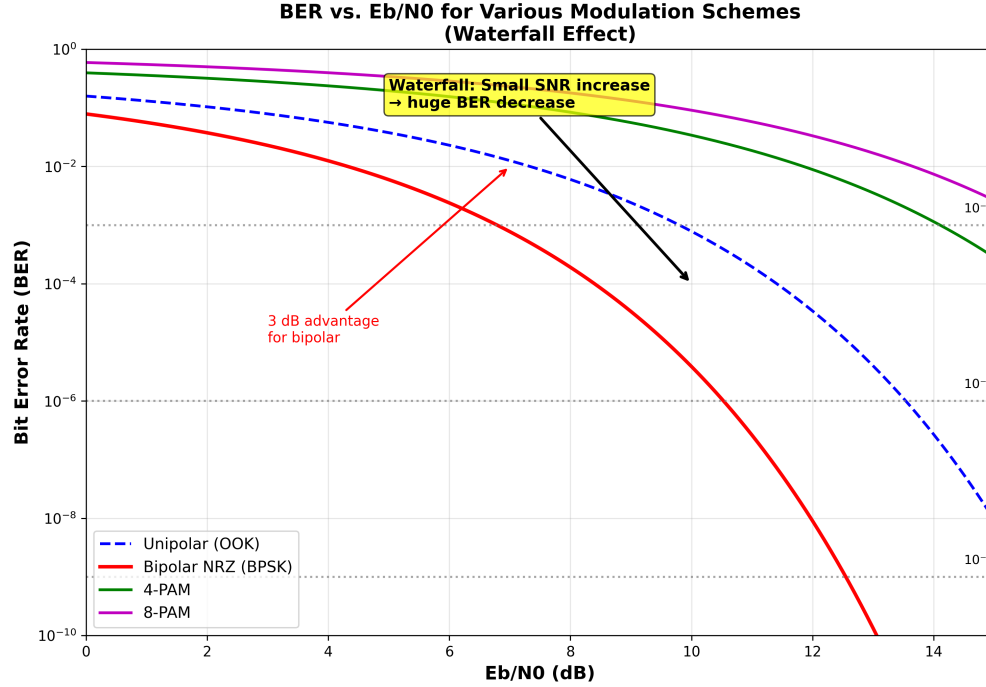


Figure 3: BER waterfall curves for different modulation schemes. Notice the dramatic decrease in error rate with increasing E_b/N_0 , and the 3 dB performance difference between unipolar and bipolar signaling.

E_b/N_0 (dB)	$Q(\sqrt{2E_b/N_0})$	BER
6 dB	$Q(2)$	2.3×10^{-2}
8 dB	$Q(2.52)$	6×10^{-3}
10 dB	$Q(3.16)$	8×10^{-4}
12 dB	$Q(4)$	3×10^{-5}
15 dB	$Q(5.6)$	10^{-8}

Table 2: Q-function values for common E_b/N_0 ratios

5.4 The Q-Function

Important values:

Approximation for large x :

$$Q(x) \approx \frac{1}{x\sqrt{2\pi}} e^{-x^2/2} \quad (35)$$

Relation to error function:

$$Q(x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right) \quad (36)$$

6 Multi-Level Signaling

6.1 M -ary Pulse Amplitude Modulation (PAM)

Motivation: Send more bits per symbol $\rightarrow$ higher data rate

M -ary PAM: Use M different amplitude levels

- $M = 2$: Binary (1 bit/symbol)
- $M = 4$: 4-PAM (2 bits/symbol)
- $M = 8$: 8-PAM (3 bits/symbol)
- $M = 16$: 16-PAM (4 bits/symbol)

Symbol values:

$$a_k \in \{-(M-1)A, -(M-3)A, \dots, -A, +A, \dots, (M-1)A\} \quad (37)$$

with spacing $2A$ between levels.

4-PAM Mapping

- Bits 00: $a_k = -3A$
- Bits 01: $a_k = -A$
- Bits 10: $a_k = +A$
- Bits 11: $a_k = +3A$

6.2 Distance and Error Probability

Minimum distance between symbols: $d_{\min} = 2A$

Probability of symbol error:[8]

$$P_s \approx 2 \left(1 - \frac{1}{M}\right) Q \left(\frac{d_{\min}}{2\sigma} \right) \quad (38)$$

For $M = 4$:

$$P_s \approx \frac{3}{2} Q \left(\frac{A}{\sigma} \right) = \frac{3}{2} Q \left(\sqrt{\frac{E_s}{5N_0}} \right) \quad (39)$$

where E_s = average symbol energy.

Converting to BER:

- For Gray coding: $P_b \approx P_s / \log_2(M)$

Key insight: Doubling M requires ~ 6 dB more E_b/N_0 for same BER!

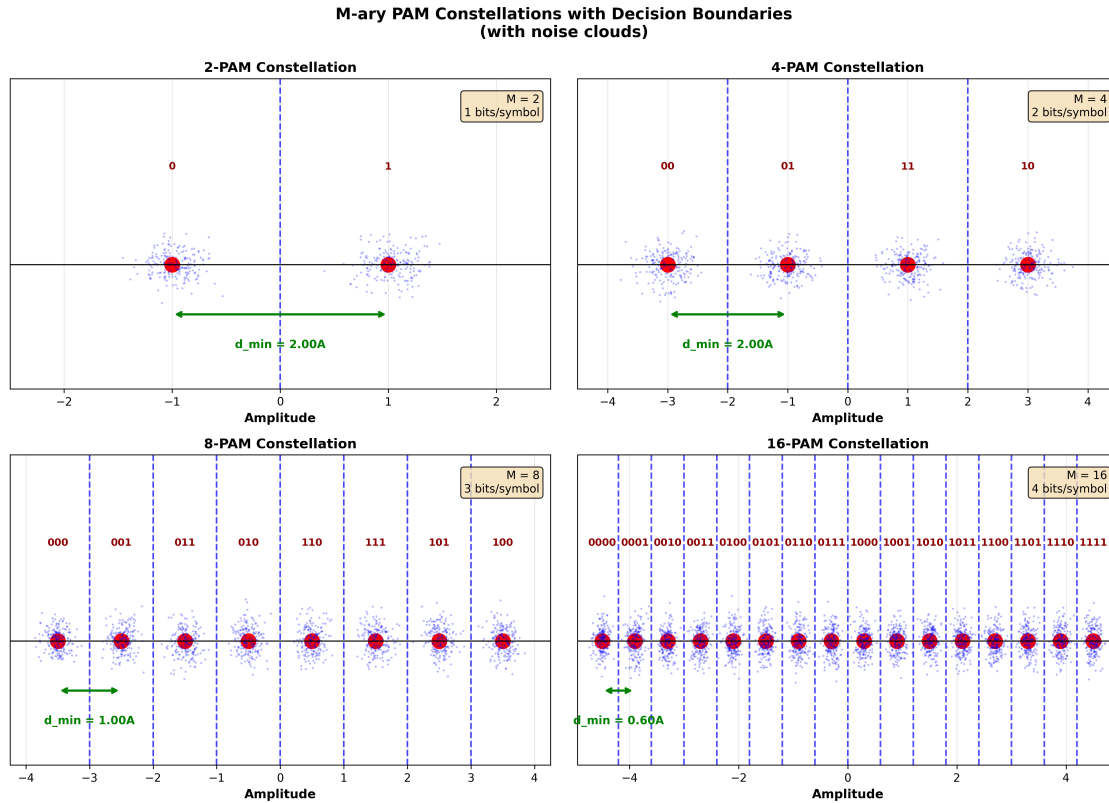


Figure 4: Constellation diagrams for M-ary PAM with $M = 2, 4, 8, 16$. As M increases, symbols are closer together, requiring higher SNR for reliable detection.

6.3 Bandwidth Efficiency

Symbol rate: $R_s = 1/T$ symbols/sec

Bit rate: $R_b = R_s \log_2(M)$ bits/sec

Bandwidth (raised cosine, $\alpha = 0.5$): $B = \frac{1.5}{2T} = 0.75R_s$

Spectral efficiency:

$$\eta = \frac{R_b}{B} = \frac{R_s \log_2(M)}{0.75R_s} = \frac{\log_2(M)}{0.75} \text{ bits/sec/Hz} \quad (40)$$

M	bits/symbol	η (bits/s/Hz)
2	1	1.33
4	2	2.67
8	3	4.0
16	4	5.33

Table 3: Spectral efficiency for different PAM orders

Higher $M \rightarrow$ Better bandwidth efficiency, but requires more power (higher SNR)

6.4 Tradeoffs: Power, Bandwidth, Error Rate, Data Rate

Fundamental Tradeoffs:

1. **Power vs. Data Rate** (fixed bandwidth):
 - Higher data rate requires more power
 - Shannon limit: $C = B \log_2(1 + \text{SNR})$
2. **Bandwidth vs. Data Rate** (fixed power):
 - Higher data rate requires more bandwidth
 - Can approach Shannon limit with coding
3. **Error Rate vs. SNR:**
 - Lower BER requires higher E_b/N_0
 - Q-function shows exponential improvement
4. **Multi-level signaling:**
 - More bits/symbol $\rightarrow$ better bandwidth efficiency
 - But requires higher SNR for same BER

Design Problem

Given: $B = 10$ MHz, $P_{\text{rx}} = -80$ dBm, $N_0 = -174$ dBm/Hz
Calculate: Maximum data rate for $\text{BER} < 10^{-6}$

Solution

1. Noise power: $N = -174 + 10 \log_{10}(10^7) = -104$ dBm
2. SNR: $-80 - (-104) = 24$ dB = 251 (linear)
3. Shannon capacity: $C = 10^7 \log_2(252) \approx 79$ Mbps
4. For $\text{BER} < 10^{-6}$: need $Q(x) < 10^{-6} \rightarrow x > 4.75$
5. Required $E_b/N_0 \approx 11$ dB
6. Achievable rate $\approx 0.7 \times C \approx 55$ Mbps (with practical coding)

7 Hands-On Activity 2: BER Calculation and Visualization

7.1 Using Python to Explore Error Rates

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import erfc

def Q(x):
```

```

    """Q-function: tail probability of standard normal"""
    return 0.5 * erfc(x / np.sqrt(2))

# Eb/N0 range (in dB)
EbN0_dB = np.arange(0, 15, 0.5)
EbN0_linear = 10**(EbN0_dB / 10)

# BER for bipolar NRZ
BER_bipolar = Q(np.sqrt(2 * EbN0_linear))

# BER for unipolar OOK (worse by 3 dB)
BER_OOK = Q(np.sqrt(EbN0_linear))

# Plot
plt.semilogy(EbN0_dB, BER_bipolar, label='Bipolar_NRZ')
plt.semilogy(EbN0_dB, BER_OOK, label='Unipolar_OOK')
plt.xlabel('Eb/N0 (dB)')
plt.ylabel('Bit Error Rate')
plt.title('BER vs Eb/N0 for Binary Signaling')
plt.grid(True, which='both')
plt.legend()
plt.ylim([1e-8, 1])
plt.show()

# Calculate required Eb/N0 for target BER
target_BER = 1e-6
# For bipolar: BER = Q(sqrt(2*Eb/N0))
# Q^(-1)(BER) = sqrt(2*Eb/N0)
from scipy.special import erfcinv
Q_inv = lambda p: np.sqrt(2) * erfcinv(2*p)
required_EbN0_linear = (Q_inv(target_BER)**2) / 2
required_EbN0_dB = 10 * np.log10(required_EbN0_linear)
print(f"For BER={target_BER}, need Eb/N0={required_EbN0_dB:.2f} dB")

```

Observe:

- Waterfall effect: small SNR increase → huge BER decrease
- 3 dB difference between bipolar and unipolar
- Typical target: $\text{BER} < 10^{-6}$ to 10^{-9}

8 Summary & Key Takeaways

8.1 What We Learned

1. Physical Media Characteristics

- Twisted pair: short distance, low cost
- Coaxial: medium performance
- Optical fiber: long distance, huge bandwidth, low loss

2. Attenuation and Link Budgets

- Use dB for power calculations
- Fiber has $\sim 100\times$ less attenuation than copper
- Link margin accounts for aging and variations

3. Shannon-Hartley Theorem

- $C = B \log_2(1 + \text{SNR})$
- Capacity grows linearly with bandwidth
- Capacity grows logarithmically with power

4. AWGN Channel Model

- $y(t) = x(t) + n(t)$
- Thermal noise sets fundamental limit
- Noise figure measures component degradation

5. Digital Signaling

- Represent bits as pulse amplitudes
- Bipolar signaling preferred (balanced, efficient)
- Matched filter maximizes SNR

6. Probability of Error

- $\text{BER} = Q(\sqrt{2E_b/N_0})$ for bipolar
- Waterfall effect: exponential improvement with SNR
- Typical target: 10^{-6} to 10^{-9}

7. Multi-Level Signaling

- More bits/symbol $\rightarrow$ better bandwidth efficiency
- But requires higher power for same BER
- Fundamental tradeoff: power $\leftrightarrow$ bandwidth $\leftrightarrow$ data rate

8.2 Connection to Networking

Why this matters:

- **Physical limits constrain network design**
 - Ethernet limited to 100m not by attenuation, but by timing
 - Fiber enables long-haul links (1000s of km)
 - Wireless has severe power constraints
- **Data rates tied to SNR and bandwidth**
 - WiFi adaptively changes modulation based on SNR
 - Higher data rates need better signal quality

- Interference reduces capacity

- **Error rates affect protocol design**

- Wired: very low BER (10^{-9}), minimal retransmission
- Wireless: higher BER (10^{-5} to 10^{-3}), need strong error correction

8.3 Next Lecture Preview

Lecture 4: Physical Layer - Wireless Communications

- Transmitter and receiver architectures
- Propagation models (path loss, fading)
- Modulation as channel selection (QAM, OFDM)
- Spectrum allocation (where WiFi, cellular live)
- Cellular reuse patterns

References

- [1] Wikipedia contributors. “Shannon–Hartley theorem.” *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Shannon-Hartley_theorem
- [2] Wikipedia contributors. “Johnson–Nyquist noise.” *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Johnson-Nyquist_noise
- [3] Wikipedia contributors. “Noise figure.” *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Noise_figure
- [4] Wikipedia contributors. “Pulse-amplitude modulation.” *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Pulse-amplitude_modulation
- [5] Wikipedia contributors. “Raised-cosine filter.” *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Raised-cosine_filter
- [6] Wikipedia contributors. “Matched filter.” *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Matched_filter
- [7] Wikipedia contributors. “Q-function.” *Wikipedia, The Free Encyclopedia*. <https://en.wikipedia.org/wiki/Q-function>
- [8] Wikipedia contributors. “Bit error rate.” *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/wiki/Bit_error_rate

Additional Resources

Textbooks:

- Proakis & Salehi, “Communication Systems Engineering”
- Haykin, “Communication Systems”
- Sklar, “Digital Communications”

Standards:

- IEEE 802.3 (Ethernet)
- TIA/EIA-568 (Structured Cabling)
- ITU-T G.652 (Optical Fiber)

Tools:

- GNU Radio (SDR platform)
- MATLAB Communications Toolbox
- Python: scipy, numpy, matplotlib